

2020; Benzell et al., 2021). The realization of general purpose technologies' full potential requires extensive co-invention (Bresnahan and Trajtenberg, 1995; Bresnahan et al., 1996, 2002; Lipsey et al., 2005; Dixon et al., 2021), a costly and time-consuming process involving the discovery of new business procedures (David, 1990; Bresnahan, 1999; Frey, 2019; Brynjolfsson et al., 2021; Feigenbaum and Gross, 2021). Consequently, many studies of machine learning technologies focus on systems-level adoption, arguing that organizational systems may require redesign to effectively take advantage of novel machine learning advancements (Bresnahan, 2019; Agrawal et al., 2021; Goldfarb et al., 2023). Appropriately designed systems can yield considerable business value and improve firm performance (Rock, 2019; Babina et al., 2021; Zolas et al., 2021), with AI tools facilitating the discovery process (Cockburn et al., 2018; Cheng et al., 2022). By employing task-level information to assess whether LLMs fulfill the criteria of a general purpose technology, we seek to merge the two perspectives for understanding the technology-labor relationship.

We attempt to build on these diverse literature streams in several ways. Echoing (Felten et al., 2023), we focus our analysis on the impact of LLMs, rather than addressing machine learning or automation technologies more broadly. Additionally, we propose a novel method that employs LLMs, specifically GPT-4, to assess tasks for exposure and automation potential, thereby bolstering human scoring efforts. Subsequently, we aggregate our findings to occupations and industries, capturing the overall potential exposure in the contemporary U.S. labor market.

3 Methods and Data Collection

3.1 Data on Activities and Tasks Performed by Occupation in the US

We use the O*NET 27.2 database (O*NET, 2023), which contains information on 1,016 occupations, including their respective Detailed Work Activities (DWAs) and tasks. A DWA is a comprehensive action that is part of completing task, such as "Study scripts to determine project requirements." A task, on the other hand, is an occupation-specific unit of work that may be associated with zero, one, or multiple DWAs. We offer a sample of tasks and DWAs in Table 1. The two datasets we use consist of:

- 19,265 tasks, consisting of a "task description" and a corresponding occupation, and
- 2,087 DWAs, where most DWAs are connected to one or more tasks, and tasks may be associated with one or more DWAs, though some tasks lack any associated DWAs.

3.2 Data on Wages, Employment, and Demographics

We obtain employment and wage data from the 2020 and 2021 Occupational Employment series provided by the Bureau of Labor Statistics. This dataset encompasses occupational titles, the number of workers in each occupation, and occupation-level employment projections for 2031, typical education required for entry in an occupation and on-the-job training required to attain competency in an occupation (BLS, 2022). We use the BLS-recommended crosswalk to O*NET (BLS, 2023b) to link the O*NET task and DWA dataset and the BLS Labor Force Demographics (BLS, 2023a), which is derived from the Current Population Survey (CPS). Both of these data sources are collected by the U.S. government and primarily capture workers who are not self-employed, are documented, and are working in the so-called formal economy.

3.3 Exposure

We present our results based on an exposure rubric, in which we define **exposure** as a measure of whether access to an LLM or LLM-powered system would reduce the time required for a human to perform a specific

or task while maintaining consistent quality. We anticipate that adoption will be highest and most immediate for applications that realize a considerable increase in productivity. Although this threshold is somewhat arbitrary, it was selected for ease of interpretation by annotators. Moreover, regardless of the chosen threshold, we guessed that the real-world reduction in task time would likely be slightly or significantly lower than our estimates, leading us to opt for a relatively high threshold. In our own validation labeling, we found that this corresponded closely to whether an LLM or LLM-powered software could perform the core part of a task or nearly the entire task.

Comparison	γ	Weighting	Agreement	Pearson's
GPT-4, Rubric 1; Human	α	E1	80.8%	0.223
	β	E1 + .5*E2	65.6%	0.591
	ζ	E1 + E2	82.1%	0.654
GPT-4, Rubric 2; Human	α	E1	81.8%	0.221
	β	E1 + .5*E2	65.6%	0.538
	ζ	E1 + E2	79.5%	0.589
GPT-4, Rubric 1; GPT-4, Rubric 2	α	E1	91.1%	0.611
	β	E1 + .5*E2	76.0%	0.705
	ζ	E1 + E2	82.4%	0.680

Table 2: Model and human comparison of agreement and Pearson's correlation scores. The agreement score is determined by looking at how often the two groups agree on the annotation (e.g. E0, E1 or E2). In the paper we use GPT-4, Rubric 1. Core tasks are given twice the weight at the occupation-level as supplemental tasks. All weights sum to one.

We then collected both human and GPT-4-generated annotations using the exposure rubric, which underlie the bulk of the analyses in this paper.

- *Human Ratings:* We obtained human annotations by applying the rubric to each O*NET Detailed Worker Activity (DWA) and a subset of all O*NET tasks and then aggregated those DWA and task scores⁵ at the task and occupation levels. The authors personally labeled a large sample of tasks and DWAs and enlisted experienced human annotators who have reviewed GPT-3, GPT-3.5 and GPT-4 outputs as part of OpenAI's alignment work (Ouyang et al., 2022).
- *GPT-4 Ratings:* We administered a similar rubric to an early version of GPT-4 (OpenAI, 2023b) but on all task/occupation pairs rather than DWAs. We made slight modifications to the rubric (which was used as a "prompt" to the model in this case) to enhance agreement with a set of human labels. Full agreement rates are given in Table 2.

We construct three primary measures for our dependent variable of interest: (i) α , corresponding to E1 in the exposure rubric above, anticipated to represent the lower bound of the proportion of exposed tasks within an occupation, (ii) β , which is the sum of E1 and 0.5*E2, where the 0.5 weight on E2 is intended to account for exposure when deploying the technology via complementary tools and applications necessitates additional investment, and (iii) ζ , the sum of E1 and E2, an upper bound of exposure that provides an assessment of maximal exposure to an LLM and LLM-powered software. We summarize agreement between annotation groups and measures in Table 2. For the remainder of the analysis, if not specified, the reader may assume that

⁵The authors annotated DWAs that clearly required a high degree of physicality or manual dexterity, and the contracted annotators labeled the remaining activities, along with a subset of tasks including those without associated DWAs and those for which there was no clear task-level annotation after aggregating the DWA annotations.